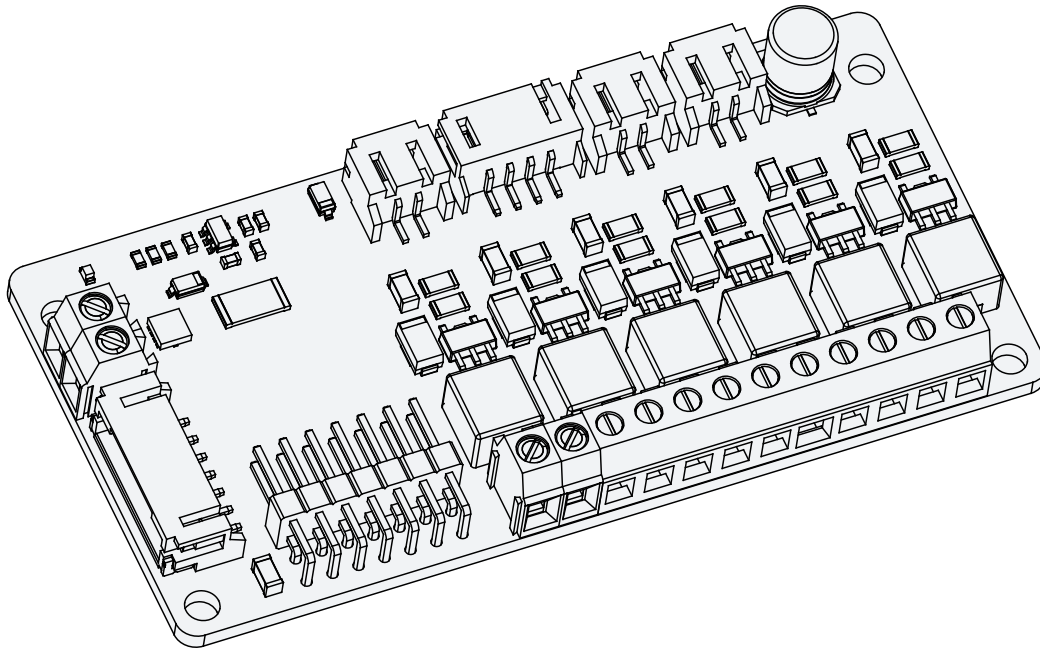




Bacopa: 6-Channel All-in-One LED Carrier/Driver



Features

- Wide input voltage range: 6 to 36 V
- Constant-current output voltage range: 2 to 32 V
- 6-channel independent 700 mA LED constant-current output
- Maximum continuous output power of 130 W (equivalent to 162 W industry rating)
- Efficiency up to 97%
- Reverse-polarity, over-temperature, and over-power protection
- 4-layer PCB with shielded integrated inductors for EMI suppression
- On-board TI INA139 high-side current sensor
- Compact credit-card form factor: 85 × 42 mm
- Easy-to-install 2.0 mm board-to-board/ribbon connectors and screw terminals
- DIN35 rail mount via accessory bracket

Applications

- High-end freshwater or marine aquarium LEDs
- Agricultural LEDs and horticultural grow lights
- Beauty and aesthetic devices
- Research and scientific LED applications
- Home lighting and ambient lighting
- Photography and studio lighting
- Other naturalistic environment lighting applications

Overview

Bacopa is the LED driver/carrier board for the Buce 6-channel WiFi LED controller (BLC06). Engineered with premium components and careful design, Bacopa delivers 700 mA constant-current output per channel, capable of driving a total of up to 162 W of LEDs (approximately 54 × 3 W high-power LED beads).

Under the control of the Buce LED controller's 6-channel phase-shifted interleaved output, 12-bit duty cycle resolution, and up to 19.5 kHz ultra-high-frequency PWM dimming signal, completely flicker-free smooth dimming is achieved while fully eliminating audible noise from inductors and ceramic capacitors.

BLM06 integrates reverse-polarity protection circuitry and a current sensor. In conjunction with the Buce controller, it provides not only up to 130 W constant-current power output but also reverse-polarity, over-temperature, and over-current protection. Combined with the companion controller's multi-segment programmable dimming, astronomical algorithm-based sunrise/sunset simulation, moonlight simulation, adaptive thermal management, and LED power monitoring, it serves as the ideal core hardware for premium aquarium luminaires, indoor agriculture/horticultural lighting, and smart lighting applications.

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1.2. Electrical Characteristics

1.2.1. Usage and Safety

- **Thermal Requirement:**
Active cooling is required when total operating power exceeds 50 W or when the voltage differential between the supply and the LED string forward voltage is large.
- **Enclosed Spaces Prohibited:**
High-load operation in enclosed enclosures without thermal management is strictly prohibited to prevent heat accumulation.
- **Power Supply Selection:**
A regulated switching power supply with output ripple less than 200 mV is recommended.
- **Hot-Plug Prohibited:**
Hot-plugging of the LED controller module or external peripherals such as fans and LEDs is strictly prohibited; failure to observe this requirement may cause equipment damage.
- **Uncontrolled Operation:**
When powered without an LED controller connected, all channels will simultaneously output maximum constant current. Ensure adequate thermal management and observe LED power ratings.
- **Series/Parallel Output Prohibited:**
This product does not employ a common-cathode or common-anode constant-current topology. Each channel may only drive an independent LED string. Electrical interconnections of any kind between output channels — including parallel, series, or cross-connections — are strictly prohibited.

1.2.2. Absolute Maximum Ratings

Operating the device beyond the absolute maximum ratings may cause permanent damage. Long-term operation at absolute maximum conditions may affect device performance and reliability.

Table 2: Absolute Maximum Ratings Table

Parameter	Symbol	Min	Max	Unit	Notes
Ambient Temperature	T_A	-25	85	°C	
Power Supply Voltage	V_{IN}	6	40	V	

1.2.3. Recommended Operating Conditions

Table 3: Recommended Operating Conditions

Parameter	Symbol	Min	Max	Unit	Notes
Ambient Temperature	T_A	-25	75	°C	
Power Supply Voltage	V_{IN}	6	36	V	$V_{IN} \geq 14$ V required for 12 V fan operation
Constant-Current Output Voltage Range	V_{OUT}	2	32	V	

1.2.4. DC Characteristics

The following parameters are measured at room temperature 25°C.

Table 4: DC Characteristics Table

Parameter	Symbol	Min	Typ.	Max	Unit	Test Conditions
Input						
Input Voltage Range	V_{IN}	2	—	32	V	
Maximum Rated Power	P_{max}	—	—	133	W	
Dimming PWM Signal Voltage Range	V_{DIM}	0.3	—	5.0	V	
Dimming PWM Signal Frequency	F_{DIM}	0.1	19.5	20	kHz	
Output						
Independent Output Channels	N_{CH}	—	6	—	—	
Per-Channel Constant Current	I_{CH}	—	0.7	—	A	
Maximum Nominal Output Power ¹	P_{LED}	—	162	—	W	
Maximum Rated Electrical Output Power	P_E	—	130	—	W	
Constant-Current Accuracy	A_C	—	3	—	%	
Rated Maximum Fan Drive Power	P_{fan}	—	6	—	W	Only one fan port may be used at a time
Output Constant-Current Accuracy	A_{oi}	—	3	—	%	
Thermal						
LED Driver Hardware Shutdown Temperature	T_{SD}	—	155	—	°C	
Efficiency	E_{max}	80	—	95	%	Varies with LED forward voltage and input voltage
Driver IC Thermal Resistance	θ_{JA}	—	45	—	°C/W	

1.2.5. Dimming Logic Levels and Control States

- **High Level (ON State):**

When the dimming signal voltage V_{DIM_H} is between 2.5 V and 5.0 V, the corresponding channel driver IC is fully enabled. At 100% duty cycle, the channel outputs the nominal maximum constant current of 700 mA.

- **Low Level (OFF State):**

When the dimming signal voltage $V_{DIM_L} < 0.3$ V, the channel's internal power switch is fully disabled and output current is 0 mA.

¹Rated per industry convention of 3 W LEDs at 700 mA.

2. Detailed Description

2.1. Constant-Current Output and Dimming Logic

All 6 channels support independent PWM (Pulse Width Modulation) dimming; the dimming signal is applied directly to the DIM pin of the constant-current driver IC. By varying the duty cycle of the input PWM signal, linear adjustment of the output current is achieved.

The 6 constant-current driver circuits are electrically independent in their topology. Each channel has a dedicated sense resistor and a dedicated dimming input pin. The modular hot-swap Buce LED controller (BLC06MKx series) or any other compatible controller can individually enable, disable, or assign different dimming duty cycles to each of the 6 channels via PWM signals. Channels operate without mutual interference or crosstalk.

2.2. Protection Mechanisms and Reliability

2.2.1. OTP Over-Temperature Protection

- **Software LED Over-Temperature Protection:**

When used with the Buce LED controller and an NTC thermistor, fully automatic software-based LED over-temperature protection is available. When the NTC-measured temperature exceeds 45°C, the fan is activated and an adaptive algorithm adjusts fan speed to track LED temperature changes. If the temperature repeatedly exceeds 65°C, the software initiates an emergency shutdown to disable the LEDs.

- **Hardware Driver Over-Temperature Protection:**

The constant-current driver IC will automatically disable its output when the die temperature reaches 155°C.

2.2.2. Reverse-Polarity Protection

An on-board ultra-low $R_{DS(ON)}$ P-channel MOSFET provides hardware reverse-polarity protection. Incorrect power supply polarity will not cause physical damage.

2.2.3. LED Open-Circuit and Short-Circuit Protection

When no LED load is connected or an open-circuit condition exists at the output, the constant-current driver IC automatically ceases switching. Since no current flows through the sense resistor (R_{CS}), the IC's power switch remains disabled, preventing damage to the driver board and surrounding components. The system resumes normal operation automatically once the open-circuit fault is cleared.

In the event of a short-circuit across any constant-current channel output (LED+ to LED-), each channel's driver IC enters a steady-state short-circuit protection mode via the on-board inductor and internal current sense resistor. Internal over-current protection and a frequency dithering mechanism limit input power, ensuring neither the driver board nor the upstream power supply is damaged by overload.

2.3. Compatible LED Controller

This module is designed for use with the companion Buce (BLC06MKx series) controller. Refer to the Buce controller documentation and datasheet for details: docs.borneoiot.com

2.4. Mobile App

This device ships with an open-source mobile app as the end-user interface. For details, refer to the online documentation: www.borneoiot.com/app.

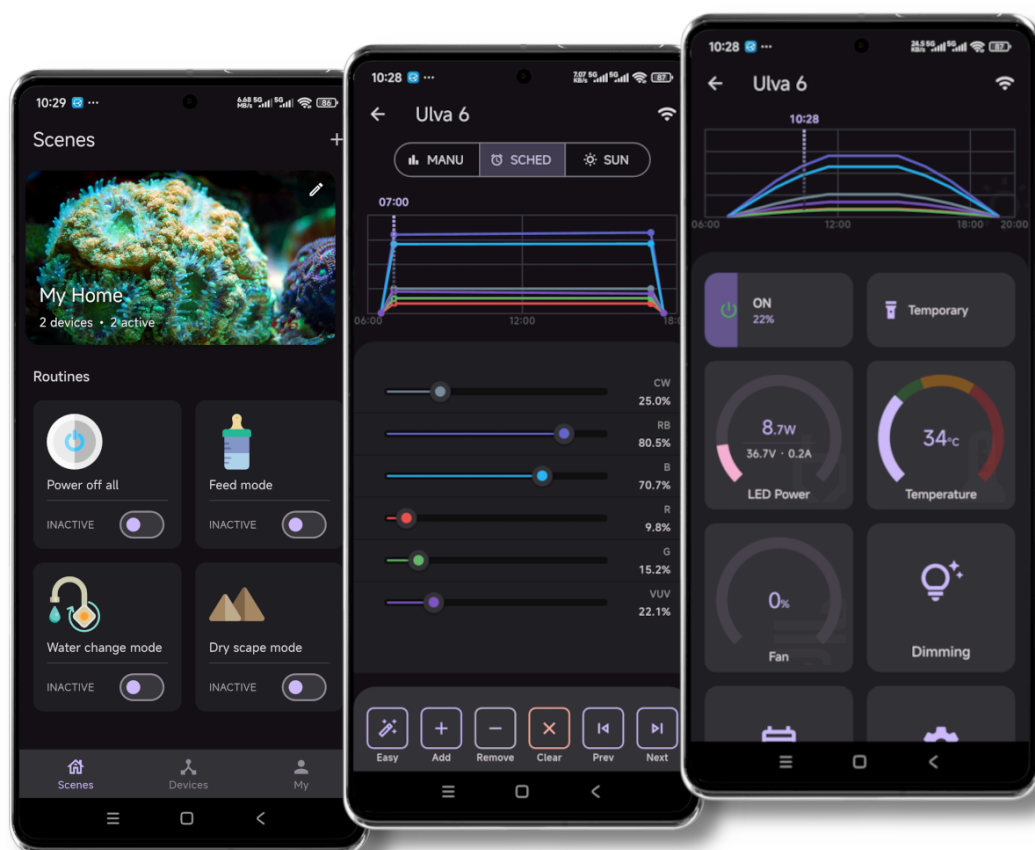


Figure 2: Mobile App Screenshots

3. Applications and Implementation

3.1. Typical Application

Figure 3 demonstrates a typical 6-channel LED lighting peripheral circuit.

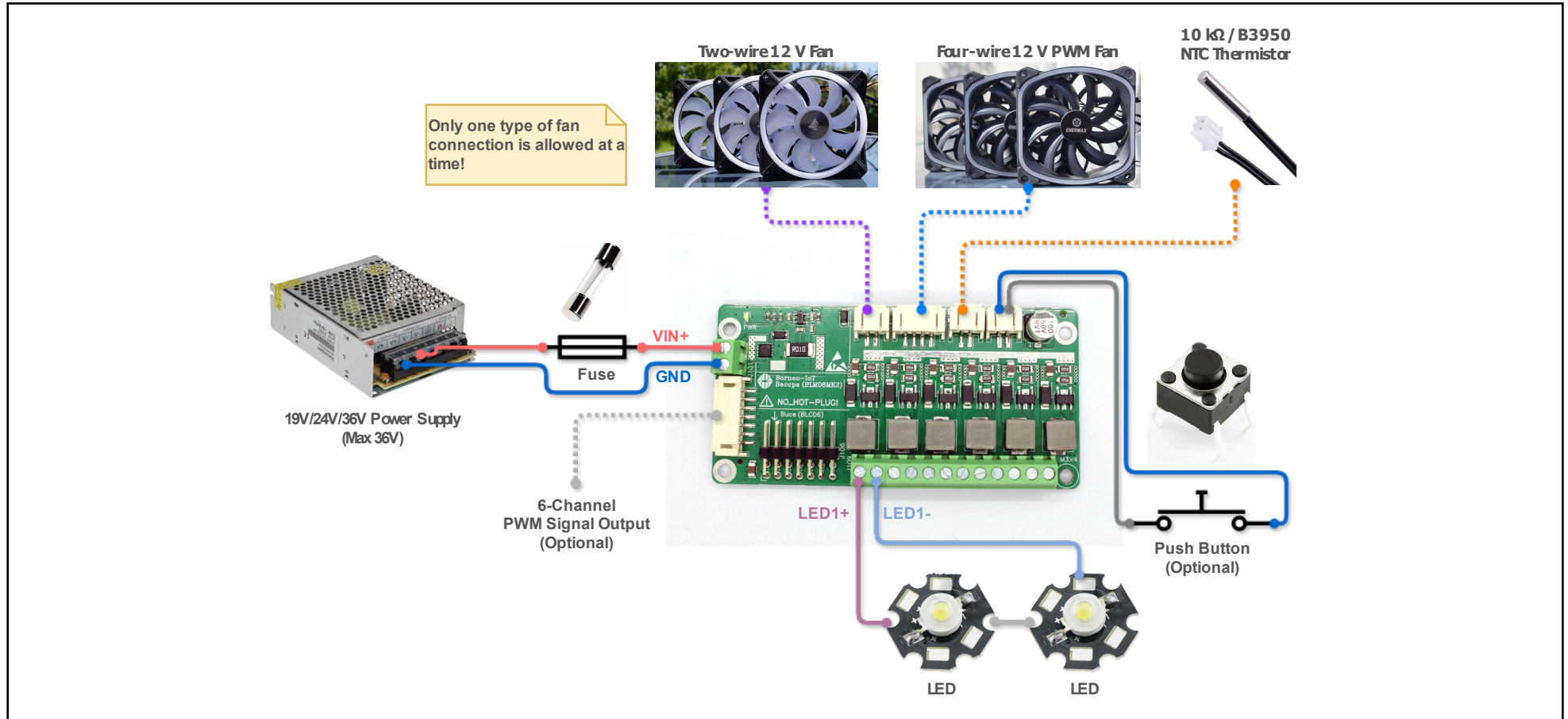


Figure 3: Simplified External Circuit Connection Diagram



3.2. Recommended Peripheral Circuits

3.2.1. Fuse

Depending on LED power and supply voltage, it is strongly recommended to install an appropriately rated fuse on the main positive power input rail as a safety measure.

3.2.2. Fan Selection

Any 12 V cooling fan rated up to 6 W may be used. Both 2-wire and 4-wire PWM fans are supported. Two-wire fans are speed-controlled via the LED controller's built-in voltage regulation circuit; 4-wire PWM fans receive a PWM speed-control signal from the LED controller.

Note

This product does not support fan tachometer monitoring. The two fan ports must not be used simultaneously.

3.2.3. NTC Thermistor Selection

Only 10 k Ω , B-value 3950 NTC resistors are supported. A tolerance of 1% or better is recommended.

The NTC should be mounted on the LED metal-core PCB or heatsink as close to the LED beads as possible. Care must be taken to keep the NTC away from cooling fan airflow to prevent erroneous temperature readings.

3.2.4. Button Selection

Any normally-open (NO) momentary push button may be used.

3.3. Total Power Limitation and Power Supply Selection

Although the driver ICs are electrically independent, all 6 channels share a common power input terminal (V_{IN+} and GND), subjecting the system to limits imposed by total current capacity and the physical current rating of the screw terminals:

1. Maximum Output Current:

When all 6 channels are simultaneously active at 100% duty cycle (700 mA each), the total constant-current output is:

$$6 \times 700 \text{ mA} = 4.2 \text{ A}$$

2. Input Current Constraint:

The rated output current of the power supply must satisfy:

$$I_{\text{Supply}} \geq \frac{V_{\text{OUT}} \times 4.2 \text{ A}}{V_{\text{IN}} \times \eta} \quad (\eta = \text{efficiency, estimated at 0.9})$$

Regardless of supply or load voltage, the total input current must not exceed 5 A due to the physical limits of the power input terminals and PCB main traces. If total load power is excessive, increase the input voltage (e.g., from 24 V to 30 V) to reduce input current.

3.4. Variable-Frequency Dimming Control and Physical Trade-offs

This product supports dynamically configurable PWM dimming frequency via firmware/app (hardware supports up to 20 kHz). Different operating frequencies present a physical trade-off between dimming depth (low-brightness linearity) and acoustic noise (inductor whine):

Frequency Range	Acoustic Behavior (Inductor Noise)	Use Cases and Trade-offs
Low range (100 Hz – 1 kHz)	Potential audible whine Falls within the human hearing sensitivity range; inductors and ceramic capacitors are prone to physical resonance.	Deep dimming applications: Highly linear dimming with extremely low minimum brightness. Suitable for noise-insensitive applications or those with acoustically isolated enclosures.
Mid range (1 kHz – 10 kHz)	Moderate mild whine Pitch increases; noise varies with load current and voltage differential.	Transitional balance range: Good dimming linearity. Not recommended for acoustically sensitive environments (e.g., bedrooms).

High range (15 kHz – 20 kHz) [Default/ Recommended]	Completely silent (inaudible) Frequency approaches or exceeds the upper limit of human hearing, eliminating acoustic vibration noise.	Premium lighting/professional aquarium: Completely silent and flicker-free under camera. Driver IC heat generation increases; better external thermal management is required.
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3.5. Thermal Design and Management

This product employs 6 independent step-down constant-current circuits. Per-channel power dissipation is primarily composed of IC quiescent power consumption, internal MOSFET conduction losses, and switching losses.

The conduction and switching losses of the internal MOSFET are closely related to the input-to-output voltage differential and operating frequency. Therefore, total board heat dissipation cannot be estimated solely from output power; it must be evaluated in conjunction with actual operating voltages.

3.5.1. Single-Channel Constant-Current Driver Power Dissipation

Single-channel constant-current driver power dissipation can be approximated as:

$$P_D \approx I_{OUT}^2 \times R_{DS(ON)} \times D + V_{IN} \times I_Q + P_{SW}$$

- I_{OUT} : Output current, fixed at 700 mA for this product.
- $R_{DS(ON)}$: Internal MOSFET on-resistance, typically 0.4 Ω .
- D : Duty cycle, where $D \approx \frac{V_{OUT}}{V_{IN}}$.
- I_Q : IC quiescent current.
- P_{SW} : Switching losses, which increase significantly with higher V_{IN} and switching frequency.

Key conclusions:

1. **Low voltage differential:**
When V_{OUT} approaches V_{IN} , duty cycle D approaches 1; losses are dominated by conduction resistance, and per-channel IC dissipation is low ($0.7^2 \times 0.5 \approx 0.245W$).
2. **High voltage differential:**
When V_{IN} is significantly higher than V_{OUT} , although duty cycle D decreases, inductor current ripple increases and switching losses P_{SW} rise sharply under high voltage, significantly increasing heat generation.

For engineering reference, the following table lists the estimated total board heat dissipation for common input voltage and load (LED string count) combinations at full load (all 6 channels active, 700 mA each)²:

Table 5: Typical Full-Load Heat Dissipation Reference Table

Input Voltage (V_{IN})	Load	Output Voltage (V_{OUT})	Differential ($V_{IN} - V_{OUT}$)	Total Heat Dissipation (Ref.)	Thermal Recommendation
24 V	6 × 3.2 V LEDs in series	~19.2 V	~4.8 V	~1.8 – 2.5 W	Low differential; natural convection or PCB copper heat spreading is sufficient
24 V	3 × 3.2 V LEDs in series	~9.6 V	~14.4 V	~3.5 – 4.5 W	Active forced-air cooling or thermal interface material required
30 V	3 × 3.2 V LEDs in series	~9.6 V	~20.4 V	~5.0 – 6.5 W	Adequate thermal management required; risk of triggering over-temperature protection

3.5.2. Thermal Design Notes

1. **Optimize Voltage Differential:**
To reduce heat generation and improve conversion efficiency, it is recommended to adjust V_{IN} such that the

²Values include typical peripheral losses from freewheeling diodes and inductors; actual data may vary from final product.



differential between V_{IN} and the LED string forward voltage V_{OUT} is 3 V to 5 V. Driving a small number of LEDs from a high supply voltage is not recommended.

2. Over-Temperature Protection Derating:

The constant-current output IC features an internal 155°C thermal shutdown. When ambient temperature is elevated or large voltage differentials cause the IC junction temperature to reach 155°C, the IC will automatically disable its output. To ensure product longevity, the PCB surface temperature should not exceed 85°C during continuous long-term operation.

3.6. Wiring and EMC Guidelines

The underlying hardware employs a high-density 4-layer PCB and fully shielded integrated inductors for EMI suppression, keeping single-board electromagnetic emissions within a low range. However, during system-level assembly, high-frequency high-current output cables may become radiating elements. To ensure long-term stable operation and eliminate inter-channel interference, the following wiring guidelines must be observed:

1. Output cables from each channel to LED loads (CH_{X+} and CH_{X-}), if longer than 30 cm, should use twisted-pair wire. Opposing current directions and opposing magnetic fields in twisted pairs greatly suppress differential-mode radiated emissions and prevent LED wiring from interfering with nearby wireless devices.
2. Constant-current output cables routed inside an enclosure must not lie against or cross the power inductor surfaces on the carrier board, and must maintain a physical clearance of at least 3 cm from the wireless LED controller's antenna to prevent high-frequency switching transients from coupling into the RF front-end and causing communication loss.
3. The PCB M3 mounting holes are directly connected to main GND. When using a metal enclosure (e.g., aluminum extrusion), ensure the carrier board establishes a good potential connection to the enclosure via the mounting screws. If the upstream power supply environment is poor (no protective earth and significant induced floating potential), install nylon washers at the mounting holes for electrical isolation to prevent external stray common-mode currents from causing faint LED flicker.
4. Install a high-frequency ferrite core (bead) over the main power input lines (V_{IN+} / GND) at the terminal block to suppress conducted common-mode noise from the upstream power supply.
5. Internal wiring within the luminaire should use high-temperature silicone or PTFE (Teflon) cables to ensure reliable insulation.

4. Mechanical Dimensions and Package Information

4.1. Mechanical Dimensions

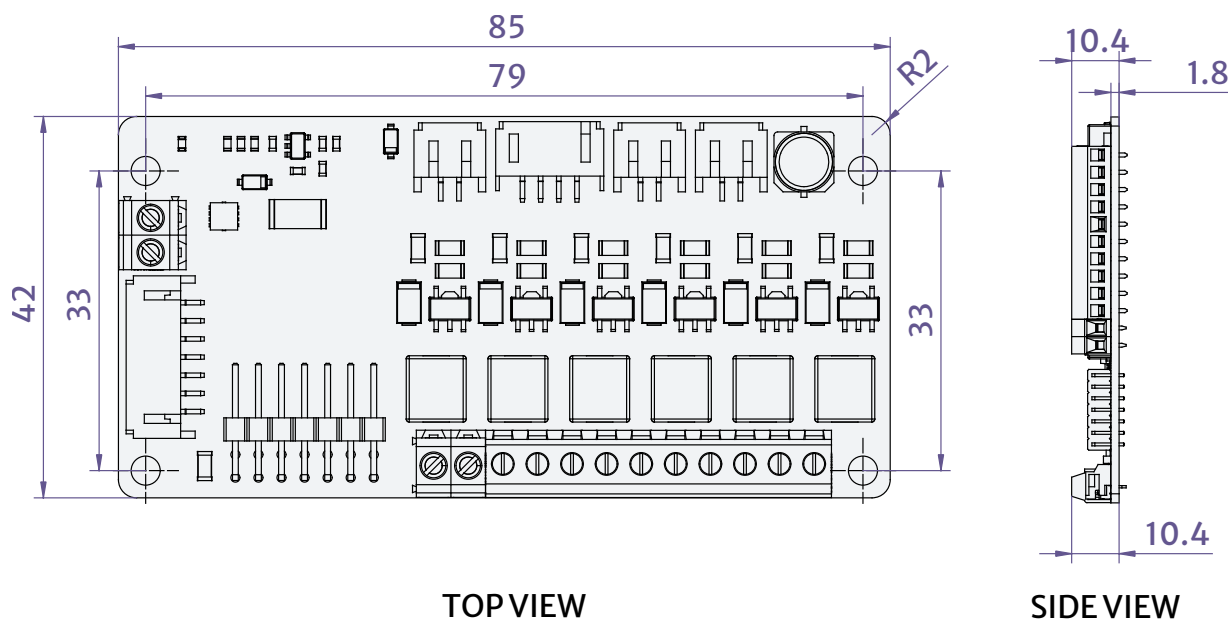


Figure 4: Package Mechanical Dimensions (Unit: mm)

4.2. Net Weight

- 32.5 g (without Buce module)
- 37.7 g (including Buce module)

5. Related Resources

- Official Website
www.borneoiot.com
- Online Documentation
docs.borneoiot.com
- GitHub Open-Source Project
github.com/oldrev/borneo
- Open-Source 6-Channel Aquarium Light Reference Design
github.com/oldrev/borneo/tree/master/hw/blb0657f



6. Indexing

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7. Revisions

REV1 - 2026-05-10
Initial version.

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